

23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : D8

Team Members :

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1. Problem Title :

Student Complaint Management System (ResolveIt)

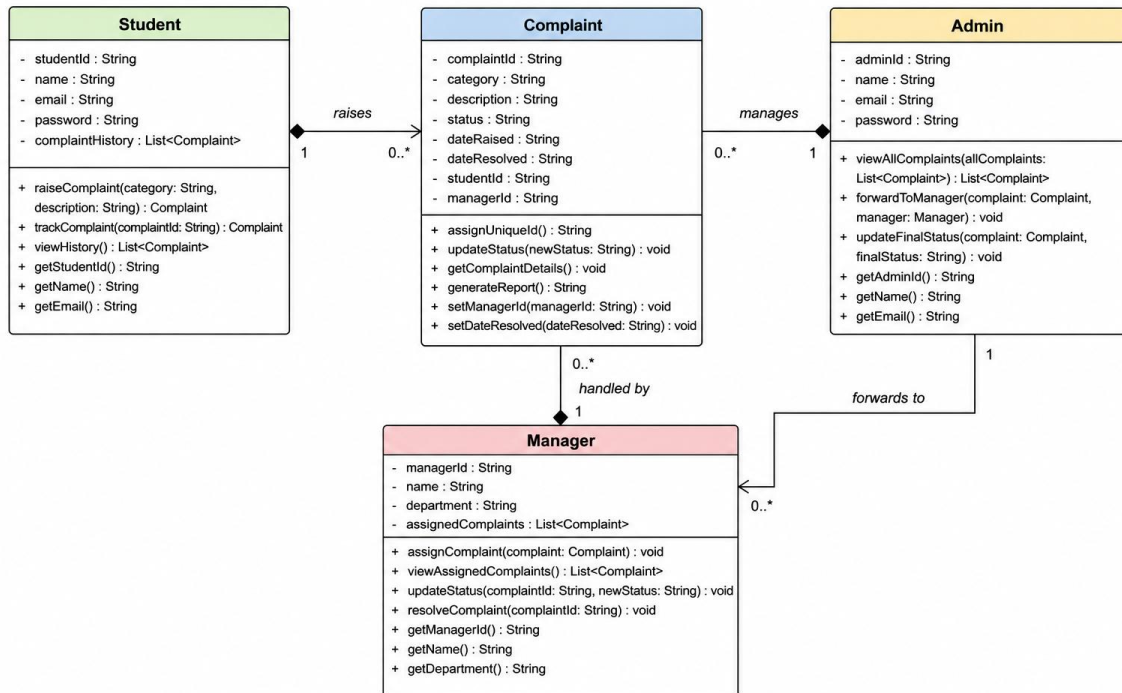
ResolveIt is a digital complaint management system developed for educational institutions to streamline the complaint registration and resolution process. It enables students to raise complaints, administrators to manage and assign them, and managers to resolve them efficiently. The system ensures transparency, accountability, and proper complaint tracking.

STRUCTURAL DESIGN

2. Class Diagram

The ResolveIt system consists of four main classes: Student, Complaint, Admin, and Manager. A Student can raise and track complaints. A Complaint stores complaint details and status. An Admin manages all complaints and forwards them to managers. A Manager handles assigned complaints and resolves them.

CLASS DIAGRAM – Resolvelt (Student Complaint Management System)



BEHAVIOURAL DESIGN

3. Use-case Diagram

a. List of actors

Student

Admin

Manager

b. List of use-cases

1. Raise Complaint

The student submits a complaint by entering complaint category and description. The system stores the complaint and generates a complaint ID.

2. Track Complaint

The student checks the current complaint status using the complaint ID.

3. View Complaint History

The student views all previously raised complaints.

4. View All Complaints

The admin views all complaints available in the system.

5. Forward Complaint to Manager

The admin assigns a complaint to a specific manager for resolution.

6. View Assigned Complaints

The manager views complaints assigned to them.

7. Update Complaint Status

The manager updates complaint progress such as 'In Progress'.

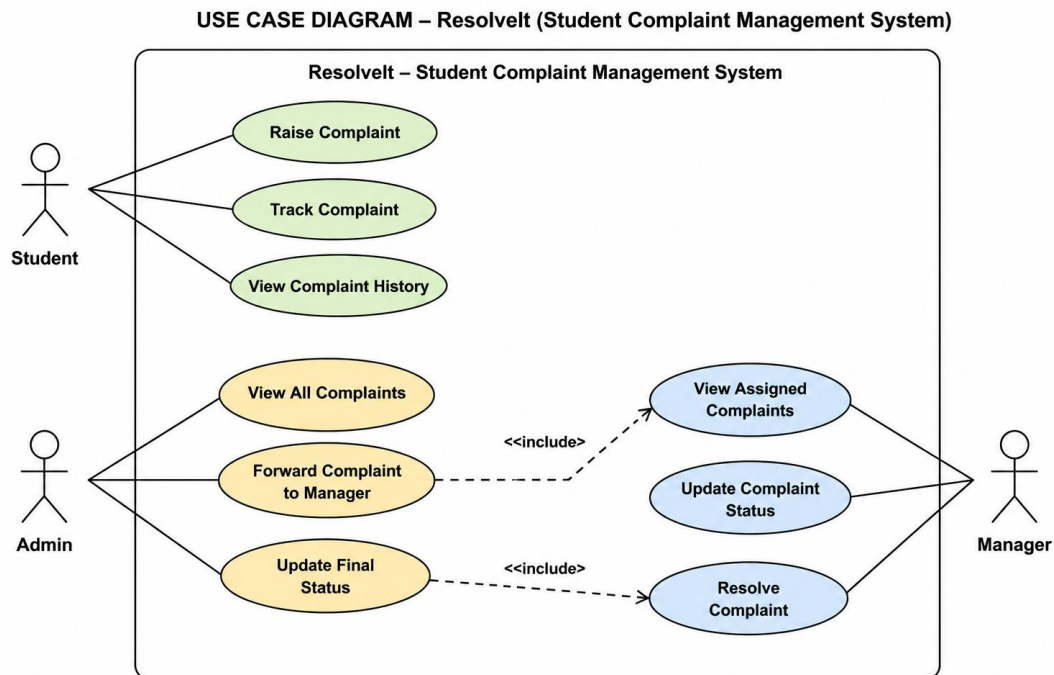
8. Resolve Complaint

The manager resolves the complaint and updates status to 'Resolved'.

9. Update Final Status

The admin confirms complaint closure after resolution.

c. Use Case Diagram (Use UML notations)

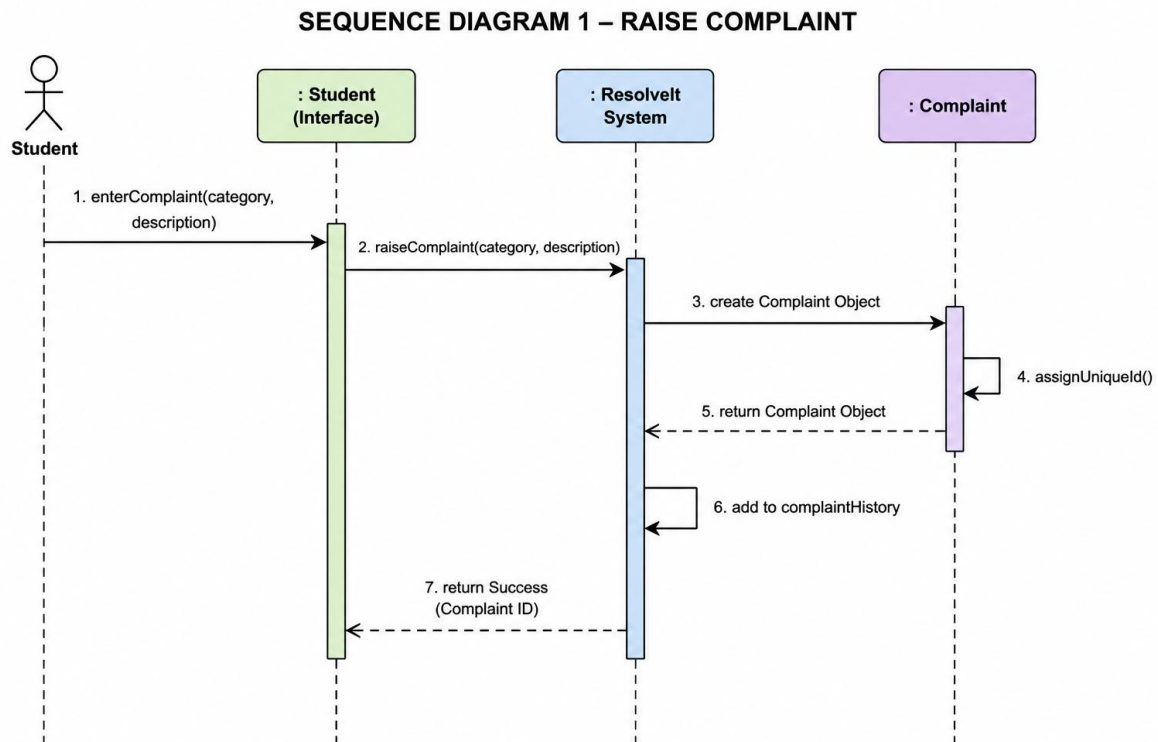


4. Sequence Diagram (Use UML notations)

a. Sequence Diagram 1 – Raise Complaint

Scenario Description: In this scenario, a student raises a complaint by entering the complaint category and description. The system creates a complaint object, generates a unique complaint ID, stores the complaint in the student's complaint history, and confirms successful submission.

Objects involved: Student, ResolveIt System, Complaint



b. Sequence Diagram 2 – Admin Forwards Complaint to Manager

Scenario Description: In this scenario, the admin reviews all complaints in the system and forwards a selected complaint to the appropriate manager. The manager receives the complaint, and the complaint status is updated to 'In Progress'.

Objects involved: Admin, ResolveIt System, Manager, Complaint

SEQUENCE DIAGRAM 2 – ADMIN FORWARDS COMPLAINT TO MANAGER

